

AIM :

To develop a programmable timed alert system using Arduino that activates both visual (LED) and auditory (buzzer) indicators at defined intervals and to analyze the timing precision using delay-based and interrupt-based approaches.

COMPONENTS REQUIRED :

• Arduino Uno • LED • 220 Ω Resistor • Piezo Buzzer • Breadboard • Jumper Wires • USB Cable

THEORY :

Arduino UNO:

Arduino Uno is a microcontroller board based on the ATmega328P. It contains digital input/output pins that can be programmed to control external electronic devices such as LEDs and buzzers. It operates at 5V and is programmed using the Arduino IDE.

LED (Light Emitting Diode):

An LED is a semiconductor device that emits light when electric current passes through it. It is used as a visual indicator in embedded systems. A 220 Ω resistor is connected in series with the LED to limit current and protect the component.

Buzzer:

A buzzer is an electronic audio signaling device used to produce sound alerts. It converts electrical signals into audible tones and is commonly used in alarm and notification systems.

Delay-Based Timing:

The `delay()` function pauses program execution for a specified time interval. It is simple to implement but blocks other operations during the delay period, reducing multitasking capability and timing efficiency.

Interrupt-Based Timing:

Interrupt-based timing uses hardware timers to execute tasks at precise intervals without stopping the processor. This approach provides better timing accuracy and is more suitable for real-time systems.

PROCEDURE:

- 1 Connect the anode (long leg) of the LED to digital pin 12 of Arduino through a 220Ω resistor.
- 2 Connect the cathode (short leg) of the LED to GND.
- 3 Connect the positive terminal of the buzzer to digital pin 9.
- 4 Connect the negative terminal of the buzzer to GND.
- 5 Place the components on the breadboard and ensure proper connections.
- 6 Connect Arduino Uno to the computer using a USB cable.
- 7 Open Arduino IDE and write the program.
- 8 Upload the code to the Arduino board.
- 9 Observe the LED glowing and buzzer sound at fixed intervals.

CIRCUIT DIAGRAM:

- LED anode → 220Ω resistor → Pin 12
 - LED cathode → GND
 - Buzzer positive terminal → Pin 9
 - Buzzer negative terminal → GND
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CODE:

```
#define LED 12
#define buz 9

void setup()
{
    pinMode(LED, OUTPUT);
    pinMode(buz, OUTPUT);
}

void loop()
{
    digitalWrite(LED, HIGH);
    digitalWrite(buz, HIGH);

    delay(1000);

    digitalWrite(LED, LOW);
    digitalWrite(buz, LOW);

    delay(2000);
}
```

EXPECTED OUTPUT:

- The LED glows and the buzzer produces sound for 1 second.
- Both the LED and buzzer remain OFF for 2 seconds.
- The process repeats continuously at fixed intervals.
- The alert system provides both visual and auditory

indications.

INFERENCE:

The programmable timed alert system using Arduino was implemented successfully using an LED and buzzer. The delay-based approach generated alerts at fixed intervals but blocked the processor during execution. Interrupt-based timing methods provide higher precision, better responsiveness, and improved multitasking capability. Therefore, interrupt-based approaches are more suitable for real-time alert systems requiring accurate timing control.